

Original papers

ASGP-IDet: Temporal behaviour localisation of beef cattle in untrimmed surveillance videos

Yamin Han^a, Jie Wu^{b,1}, Qi Zhang^a, Xilong Feng^a, Yang Xu^a, Taoping Zhang^c, Bowen Wang^{d,e,*}, Hongming Zhang^{a,f}^a College of Information Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China^b Animal Science Department, The University of Tennessee, Knoxville, TN 37996, USA^c Shaanxi Kingbull Livestock Company Limited, Yangling, Shaanxi 712100, China^d College of Economics and Management, Northwest A&F University, Yangling, Shaanxi 712100, China^e Yangling Livestock Industry Innovation Center, Yangling, Shaanxi 712100, China^f Shaanxi Agricultural Information Intelligent Sensing and Analysis Engineering Technology Research Center, Yangling, Shaanxi 712100, China

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ABSTRACT

An accurate analysis of beef cattle behaviour provides valuable information about their important characteristics such as health status and fertility. Recent studies have utilised computer vision technologies to recognise beef cattle behaviour in trimmed videos with a single behaviour. However, these methods ignore the fact that surveillance videos in real farm circumstances are usually untrimmed and contain multiple behaviour instances and background scenes, which limit their applicability. To address this issue, we propose a temporal behaviour localisation method using aggregate scalable-granularity perception instance detection (ASGP-IDet) to localise beef cattle behaviours in untrimmed videos. It provides semantic information, such as “when does a specific behaviour start and end?” and “duration of a specific behaviour”. To this end, a feature pyramid with ASGP blocks was designed to aggregate information across different temporal granularities. The trident head was then employed to achieve precise behaviour boundary predictions, and the classification head was used to predict the behaviour category of the instance. Finally, a novel centre-start-end instant offset loss (CSEIO Loss) is proposed for correct offsets at the start, end, and temporal centre of behaviours. Experiments on the newly collected Cattle Temporal Action dataset demonstrated that ASGP-IDet outperformed other state-of-the-art approaches. It achieved mAP scores of 93.93%, 93.74%, 93.22%, 92.29%, and 87.46% at tIoU thresholds [0.3:0.7:0.1], specifically, an average mAP of 92.13%, and an average processing time of 92.9 ms per video. These findings introduce an efficient method for localising the temporal behaviour of beef cattle in untrimmed farm surveillance videos and further support precision livestock farming.

1. Introduction

Accurate analysis and understanding of the behaviours of beef cattle in surveillance videos are highly significant for disease prevention, fertility, and production yield of beef cattle (Wang et al., 2020; Ban and Guan, 2021; Mays et al., 2013). For example, the semantic information of mounting behaviour contained in surveillance videos, such as “when does mounting behaviour start and end?” and “duration of mounting behaviour” provides an important basis for decision-making in cattle farm management. Mining the above semantic information on behaviours in surveillance videos is a challenging but essential task in realistic farm circumstances, which can save tremendous time and

computational costs.

With the development of computer vision technologies, some researchers (Zheng and Qin, 2023; Lauer et al., 2022; Wang et al., 2023) have focused only on mining the semantic information of behaviour categories in surveillance videos. Among these cattle behaviour recognition methods, some scholars have utilised object detection algorithms (W.A.N.G. Shaohua, 2021; Alameer et al., 2022; Wang et al., 2022; Zheng and Qin, 2023) and proposed estimation technologies (Lauer et al., 2022; Li et al., 2022; Nasiri et al., 2022; Dong Lizhong et al., 2022; ZHANG Hongming et al., 2022) based on still images to recognise cattle behaviour. Unfortunately, the performance of these

* Corresponding author.

** Corresponding author at: College of Information Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China.

E-mail addresses: chinawbw@126.com (B. Wang), zhm@nwsuaf.edu.cn (H. Zhang).¹ Jie Wu contributed equally to this work.



Fig. 1. Illustration of contrast between a trimmed video for the “walking” behaviour from the adult cow datasets (Qiao et al., 2022) and an untrimmed video for the “running” and the “fighting” behaviour from the real farm circumstances. Note that the “walking” instance is marked in yellow [■], “running” instance is in orange [■] and “fighting” instance is in blue [■]. A red dashed box marks the area where the behaviour occurs. The length of each colour block represents the duration of the behaviour.

early image-based studies remains limited because they lack the ability to classify complex behaviours that are inherently temporal and/or correlated in nature. By contrast, to further improve the accuracy of behaviour recognition, other studies (Chen et al., 2020; Qiao et al., 2022; Wang et al., 2023) extracted the spatio-temporal features of videos and employed LSTM or CNN models to recognise the behaviours of cattle. For example, Qiao et al. (2022) combined a C3D (Convolutional 3D) network with a convolutional long short-term memory (ConvLSTM) to effectively fuse spatial and temporal features to enhance behaviour classification accuracy.

Despite the success of the above cattle behaviour recognition methods, as shown in Fig. 1, they only recognised the behaviour of beef cattle in a trimmed video that only contained one behaviour and ignored the fact that the surveillance videos in real farm circumstances are usually untrimmed and contain multiple behaviour instances as well as the video content of background scenes. To make these cattle behaviour recognition methods applicable to real farm circumstances, Ma et al. (2022) attempted to equip their cattle behaviour recognition method with a sliding-window mechanism on untrimmed videos. Specifically, the sliding window mechanism was first applied to an untrimmed video, and the obtained video clips were fed into their model to recognise behaviour categories. Unfortunately, the performance of this strategy on untrimmed videos was still limited by many factors, such as the hyperparameters of the sliding window size. In addition, the videos used to verify this strategy were spliced from existing trimmed videos that contained one behaviour, which differs significantly from actual surveillance videos.

Focusing on those limitations, in this work, we first collected the realistic beef cattle surveillance videos named the Cattle Temporal Action dataset, which included many disturbance factors, such as large variation in video length, collected time, the number of temporal behaviour labels, etc. Based on the dataset, we propose an aggregate scalable-granularity perception instance detection (ASGP-IDet) method to recognise and localise beef cattle behaviours within an untrimmed surveillance video sequence. Our ASGP-IDet method can mine the semantic information of behaviours in surveillance videos, such as “when does a specific behaviour start and end?” and “duration of a specific behaviour”. Video features were first extracted using a pre-trained behaviour recognition network, DB-TEAF (Han et al., 2024). Subsequently, a feature pyramid (Section 2.2.3) with aggregate scalable-granularity perception (ASGP) blocks was designed to aggregate the video features across different temporal granularities. Then, the prediction head, composed of the trident head (Shi et al., 2023) and classification head (Zhang et al., 2022a), was employed to predict the occurrence instances and behaviour category at each scale feature space. Finally, the centre–start–end instant offset loss (CSEIO Loss) was proposed to correct the offset distance of the start boundary, end boundary, and temporal centre of the behaviour. Our ASGP-IDet method produces superior results by leveraging the benefits of the proposed CSEIO Loss and traditional focal losses (Lin et al., 2017). The proposed ASGP-IDet method obtained 93.93%, 93.74%, 93.22%, 92.29%, 87.46% of mean average precision (mAP) with a threshold interval of 0.1 ranging from 0.3 to 0.7 and 92.13% overall average mAP. For each video, the average processing time of the proposed

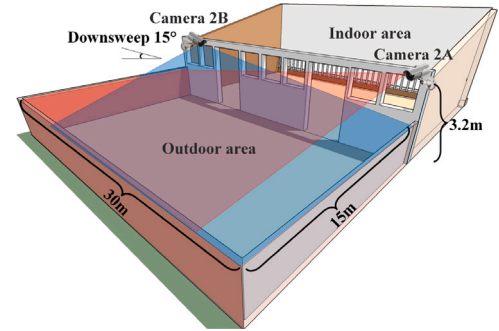


Fig. 2. Sketch map of cattle pen monitoring.

method is 0.0929 s.

The main contributions of our work can be summarised as follows: (1) We propose the ASGP-IDet method to recognise and localise beef cattle behaviours within an untrimmed surveillance video sequence. It not only provides semantic information but also saves tremendous time and computational costs for analysing surveillance videos. (2) A feature pyramid with Aggregate Scalable-Granularity Perception (ASGP) blocks is designed to aggregate video features across different temporal granularities. The novel CSEIO Loss is proposed to achieve high temporal localisation accuracy by correcting the offset distance of the start boundary, end boundary, and temporal centre of the behaviour instance. (3) The Cattle Temporal Action dataset with challenging factors of a large variation in video length, collection time, and number of temporal behaviour labels is introduced for open benchmark verification. Experimental results on the Cattle Temporal Action dataset showed that ASGP-IDet achieved the most competitive performance in beef cattle temporal behaviour localisation.

2. Material and methods

2.1. Dataset

The surveillance videos used in this study were recorded in the outdoor area of a beef cattle farm using original monitoring equipment during the day and night at Ningxia Yinchuan Benwang Ecological Breeding Company from 21 to 29 July, 2019. The participants in our study were Simmental and Angus cattle. As shown in Fig. 2, the outdoor area of the beef cattle farm was 30 m × 15 m (length × width). Two monitoring cameras (HIKVISION DS-IPC-T12H-1a) were installed on the support wall of the cowshed at a height of 3.2 m and kept at an inclination of approximately 15° downward to capture most beef cattle in the outdoor area.

Inspired by THUMOS14 (Jiang et al., 2014) for the human temporal action localisation task, we built a beef cattle temporal behaviour localisation dataset called Cattle Temporal Action. The dataset included 1337 videos, of which 1067 were used for training and 270 were used for testing. Each video contained an average of 1.49 behaviour instances with temporal labels including the starting instant, ending

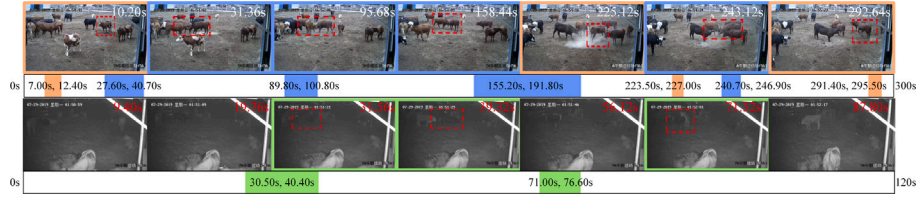


Fig. 3. Video examples in Cattle Temporal Action. A red dashed box marks the area where the behaviour occurs. Green [■] represents mounting behaviour, orange [■] indicates running behaviour, and blue [■] is fighting behaviour. The length of each colour block represents the duration of the behaviour.

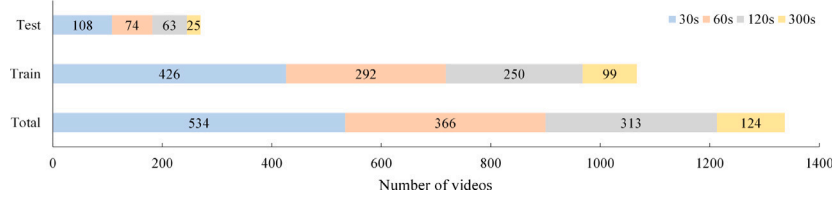


Fig. 4. Statistics of videos with different lengths.

Table 1

Statistics of videos with different numbers of instances.

Different numbers of instances	1	2	3	4	5	6	7	8
Number of videos	923	270	96	26	10	5	5	2

instant, and behaviour category. In this study, we focused on the mounting, running, and fighting behaviours of beef cattle, because these behaviours are highly indicative of their health status, fertility, and production yield. Examples from this dataset are shown in Fig. 3. In the first video with a total duration of 300 s, the beef cattle were observed running at three time periods (i.g., 7.00–12.40 s, 223.50–227.00 s, and 291.40–295.50 s), and fighting at four time periods (i.g., 27.60–40.70 s, 89.80–100.80 s, 155.20–191.80 s, and 240.70–246.90 s). In the second video, with a total duration of 120 s, mounting behaviours were recorded at two time intervals (i.g., 30.50–40.40 s and 71.00–76.60 s). The dataset has many challenging factors, such as large variations in video length, collection time, the number of temporal behaviour labels, etc. As shown in Fig. 4, the length of these videos varies with 534 videos being 30 s long, 366 videos being 60 s, 313 videos being 120 s and 124 videos being 300 s. As shown in Table 1, the videos in our dataset contained different numbers of behaviour instances.

2.2. Aggregate scalable-granularity perception instance detection (ASGP-IDet) method

Problem definition. Similar to the definition of the human temporal action localisation methods (Shi et al., 2023; Shou et al., 2016; Tang et al., 2023; Zhang et al., 2022a), we clarified the purpose of the beef cattle temporal behaviour localisation task. Assume that there is a set of untrimmed beef cattle behaviour videos $D = \{V_i\}_{i=1}^n$, where each video V_i may vary in the number of instances T_i . Specifically, given an untrimmed beef cattle behaviour video V_i , we can obtain the features $x_i \in R^{2048 \times T_i}$ by using the pre-trained behaviour recognition network DB-TEAF (Han et al., 2024). The beef cattle temporal behaviour localisation task aims to predict the temporal behaviour instance set $\Psi_i = \{\psi_i^1, \psi_i^2, \dots, \psi_i^M\}$ based on the input feature x_i , where M is the total number of behaviour instances in video V_i . Each instance ψ_i^m consists of (s_m, e_m, r_m) , where $s_m \in [0, T_i]$ is the start instant, $e_m \in (s_m, T_i]$ is the end instant, r_m indicates the beef cattle behaviour category of ψ_i^m .

2.2.1. Overview

Fig. 5 shows the structure of the proposed ASGP-IDet method. There are three main parts of the ASGP-IDet method: a pre-trained behaviour

recognition network DB-TEAF (Han et al., 2024), an ASGP feature pyramid module, and a prediction head. When an untrimmed video V_i arrives, video features x_i are extracted using the pre-trained behaviour recognition network. Subsequently, an ASGP feature pyramid module was built to obtain the temporal features of different scales based on x_i . Finally, the features of each scale obtained by the pyramid are sent to the prediction head, which is composed of classification and trident heads, to detect all behaviour instances with the start, end, and behaviour categories. Finally, the soft-NMS (Bodla et al., 2017) operator was applied to the predicted behaviour instances from all scales to eliminate duplicates and obtain the final results. Our ASGP-IDet method can produce superior results by leveraging the benefits of the proposed CSEIO Loss and traditional focal loss.

2.2.2. Pre-trained behaviour recognition network

DB-TEAF, a behaviour recognition network for beef cattle, is utilised to extract features from video data. It is a dual-branch network comprising a temporal excitation and aggregation branch, and a spatial branch. The temporal excitation and aggregation branch captures both long- and short-term temporal features, whereas the spatial branch extracts spatial features. Given an input video V_i , the video features $x_i \in R^{2048 \times T_i}$ are extracted from the last convolutional layer of DB-TEAF. Further details of DB-TEAF can be found in the original study (Han et al., 2024).

2.2.3. ASGP feature pyramid

As illustrated in Fig. 5, when the video features x_i extracted from the pre-trained behaviour classification network (Han et al., 2024) arrive, the ASGP feature pyramid is built to obtain temporal features of different scales based on x_i . In particular, the ASGP feature pyramid contains six layers, each consisting of an ASGP Block and a group normalisation operation (Wu and He, 2018). It should be noted that through each pyramid layer, the number of channels in x_i is compressed to half of the previous one.

For convenience, we consider the calculation process of the j th layer of the feature pyramid, which consists of an ASGP Block and a group normalisation operation, as an example. The ASGP block (Fig. 6) contains three main branches: an instance-level branch, a window-level branch and a global aggregate branch. Mathematically, the output x_i^j of ASGP block in j th layer can be written as

$$x_i^j = \gamma(x_i^{j-1}) + \delta(x_i^{j-1}) + \varepsilon(x_i^{j-1}) \quad (1)$$

, where $\gamma(\cdot)$ is the output of the instance-level branch, $\delta(\cdot)$ is the output of the window-level branch, $\varepsilon(\cdot)$ is the output of the global aggregate branch, and x_i^{j-1} represents the output of ASGP block in $j - 1$ th layer.

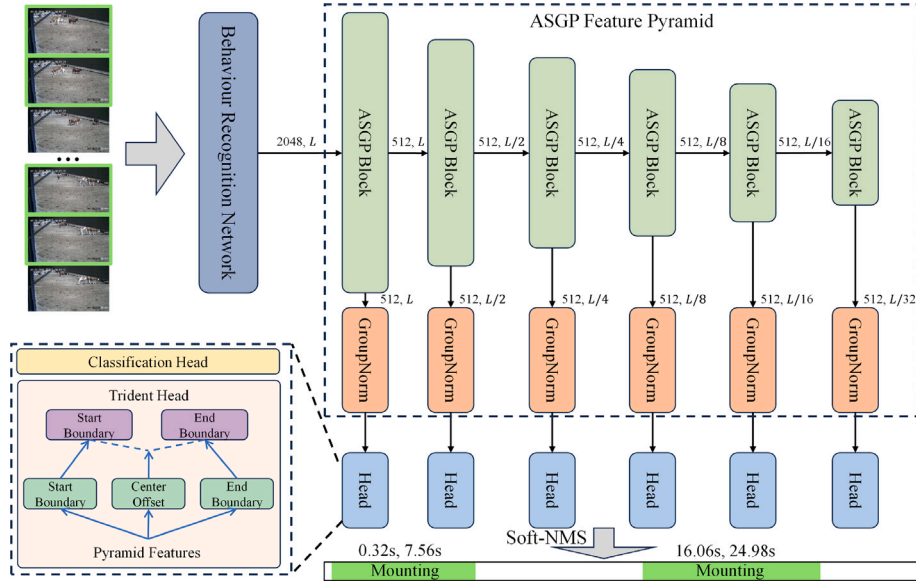


Fig. 5. Architecture of the proposed ASGP-IDet method.

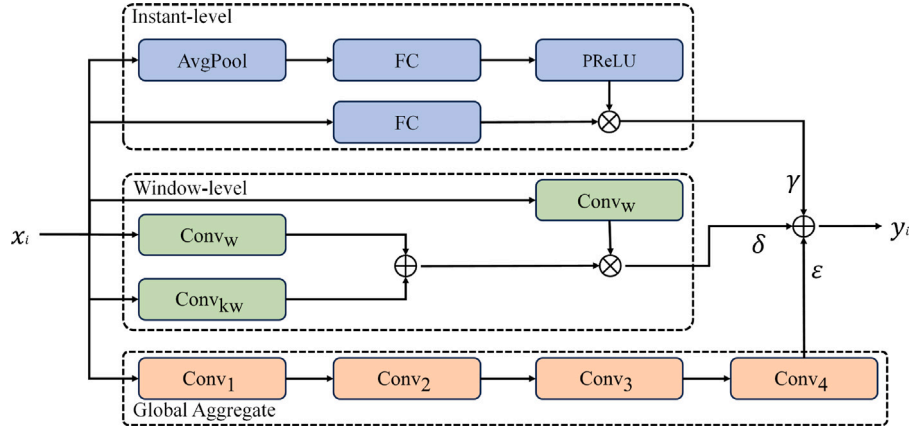


Fig. 6. The structure of ASGP block.

The instance-level branch $\gamma(x_i^j)$ is designed to increase the feature discriminability between action and non-action instances by increasing the feature distance, which is calculated as follows:

$$\gamma(x_i^{j-1}) = FC(x_i^{j-1}) \times PReLU(FC(AvgPool(x_i^{j-1}))) \quad (2)$$

where FC denotes the fully connected layer, $PReLU$ the Parametric Rectified Linear Unit (PReLU) (He et al., 2015) activation function, $AvgPool$ the average pooling operation.

The window-level branch $\delta(x_i^{j-1})$ is used to capture semantic content, which is calculated as follows:

$$\delta(x_i^{j-1}) = Conv_w(x_i^{j-1}) \times (Conv_w(x_i^{j-1}) + Conv_{kw}(x_i^{j-1})) \quad (3)$$

where $Conv_w$ denotes the one-dimensional depthwise convolution layer (Chollet, 2017) with w kernel size and $Conv_{kw}$ is the one-dimensional depthwise convolution layer with $(w+1) \times k$ kernel size.

The designed global aggregate branch was inspired by Stacked Hourglass Networks (Newell et al., 2016), which were introduced to aggregate local and global motion features. The global aggregate branch $\epsilon(x_i^{j-1})$ is defined as

$$\epsilon(x_i^{j-1}) = Conv_4(Conv_3(Conv_2(Conv_1(x_i^{j-1})))) \quad (4)$$

, where $Conv_1$ and $Conv_2$ denote the above two deep convolution layers, which reduce the dimensions of x_i^{j-1} , $Conv_3$ and $Conv_4$ indicate the last

two deep-wise convolution layers which increase the dimension of x_i^{j-1} .

Subsequently, the output x_i^j of the ASGP block is fed into a group normalisation operation (Wu and He, 2018) to obtain the final output $y_i^j \in R^{512 \times \frac{T_i}{2^{j-1}}}$ of the j th layer in the ASGP feature pyramid.

2.2.4. Prediction head

As shown in the bottom left corner of Fig. 5, the prediction head is composed of a classification head (Zhang et al., 2022a) and a trident head (Shi et al., 2023) in parallel. It utilises the features y_i^j of the j th scale from the pyramid to detect all behaviour instances $\hat{\Psi}_i = [\hat{\psi}_i^1, \hat{\psi}_i^2, \dots, \hat{\psi}_i^N]$ with the start instant s_n , end instant e_n and behaviour category r_n , where N is the total number of the prediction instances in video V_i . The parameters of the prediction head are shared across all pyramid levels to reduce the number of parameters.

Classification Head. Given feature y_i^j , the classification head predicts the probability p_n of behaviour category r_n . Specifically, the classification head comprised three layers of 1D convolutions: layer normalisation (for the first two layers), ReLU activation, and a sigmoid function. More details can be found in ActionFormer (Zhang et al., 2022a).

Trident Head. Given feature y_i^j , the trident head, including the start head, end head, and centre-offset head, is used to predict the initial start instant s_i and initial end instant e_i . First, the start and end heads

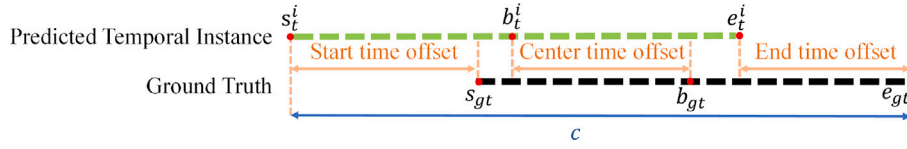


Fig. 7. The schematic diagram of CSEIO Loss calculation.

characterise the response value for each instant t as the starting or ending point of the behaviour, respectively. Second, the centre offset head estimates the start boundary distance d_{st} and the end boundary distance d_{et} of the starting and ending points. Subsequently, the initial predicted instants s_t and e_t are decoded as follows:

$$s_t = (t - d_{st}) \times 2^{j-1} \quad (5)$$

$$e_t = (t + d_{et}) \times 2^{j-1} \quad (6)$$

Finally, the predicted instant set $[s_t, e_t]$ is deduplicated by Soft-NMS to obtain the final predicted instants $[s_n, e_n]$. Note that all heads are composed of three convolutional layers, and more details can be found in TriDet (Shi et al., 2023).

2.2.5. Loss function

The overall loss $\mathcal{L}$ comprises the classification loss $\mathcal{L}_{cls}$ and the localisation loss $\mathcal{L}_{CSEIO}$, which is defined as follows:

$$\mathcal{L} = \frac{1}{N_{pos}} \sum_{i,t} 1_{\{c_t^i > 0\}} (\sigma_{tIoU} \mathcal{L}_{cls} + \mathcal{L}_{CSEIO}) + \frac{1}{N_{neg}} \sum_{i,t} 1_{\{c_t^i = 0\}} \mathcal{L}_{cls} \quad (7)$$

where N_{pos} and N_{neg} represent the numbers of positive and negative samples, respectively. c_t^i is greater than zero when the model predicts a behaviour instance in the predicted instances; otherwise, c_t^i is equal to zero. σ_{tIoU} is the temporal intersection over union (tIoU), calculated as $\sigma_{tIoU} = \frac{A \cap B}{A \cup B}$, where A is the predicted instant, and B is the real instant. $\mathcal{L}_{cls} = -\alpha_n (1 - p_n)^\gamma \log p_n$ is the classification loss calculated using the focal loss (Lin et al., 2017), where $-\alpha$ is set to 0.25, γ is set to 2.0, p_n is the class probability of model prediction.

$\mathcal{L}_{CSEIO}$ is the centre, start, and end instant offset loss (CSEIO Loss), as shown in Fig. 7, which is used to calculate the finely offset of instant based on tIoU. Mathematically, the CSEIO Loss can be written as

$$\mathcal{L}_{CSEIO} = 1 - \sigma_{tIoU} + \frac{\rho^2(s_t^i, s_{gt})}{c^2} + \frac{\rho^2(b_t^i, b_{gt})}{c^2} + \frac{\rho^2(e_t^i, e_{gt})}{c^2} \quad (8)$$

, where $\rho(\cdot)$ is the Euclidean distance, c means the length of the union of predicted instant and actual instant, s_t^i and s_{gt} denote the start instant of o_t^i and o_{gt} , b_t^i and b_{gt} denote the central points of o_t^i and o_{gt} , e_{pred} and e_{gt} denote the end instant of o_t^i and o_{gt} .

2.3. Evaluation indicators

Following the temporal action localisation task (Shi et al., 2023; Tang et al., 2023; Zhang et al., 2022a), metrics, including the mAP at tIoU thresholds and the average mAP by averaging across several tIoUs, were used for evaluation. We selected tIoU thresholds of 0.3, 0.4, 0.5, 0.6, and 0.7 in our experiment. Concurrently, we evaluated the complexity of the model using the metrics of floating point operations (FLOPs), parameters (Params), and inference time.

Specifically, tIoU is mathematically expressed as $tIoU = \frac{A \cap B}{A \cup B}$, where A is the predicted instant and B is the real instant. When the tIoU calculated by the predicted segment and ground truth segment was larger than the threshold, the predicted segment was considered a positive sample. The calculation of mAP is defined as follows:

$$mAP = \frac{\sum_{j=1}^C AP_j}{C} \quad (9)$$

, where C denotes the total number of categories; In our experiment, these parameter C was set to three, which is equal to the number of categories of behaviour in beef cattle. AP_j is the average precision (AP)

for category j , which is calculated as follows:

$$AP_j = \frac{\sum_{k=1}^n (P_k \times rel_k)}{\sum_{k=1}^n rel_k} \quad (10)$$

where n is the total number of videos, P_k is the precision at cut-off k , rel_k equals 1 if the video ranked k is a true positive and zero otherwise.

2.4. Experiment setup

In our study, the proposed ASGP-IDet was implemented using Python 3.7.13 and Pytorch 1.12.1. The experiments were conducted on a computer with Ubuntu 20.04.3, one GeForce GTX 3090 GPU, an Intel (R) Core (TM) i9-11900@2.50 GHz CPU, 128 GB memory, and 2 TB hard disk. During the training phase, the batch size was 32, the warm-up epochs were 20, and the epochs were 200. The learning rate was initialised to 0.0001 and the weight decay was 0.0001. The number of layers in the ASGP feature pyramid is six. The entire training process takes approximately 3.2 h on a GeForce GTX 3090 GPU.

3. Results and discuss

3.1. Comparisons with the state-of-the-art temporal action localisation methods

Unlike the prior methods that focus on recognising single behaviours in trimmed videos, our work proposed a novel temporal behaviour localisation task for beef cattle, designed for analysing and understanding real farm surveillance videos. Therefore, for a fair comparison, we compared the proposed method with the state-of-the-art methods in the field of temporal action localisation such as TAGS (Nag et al., 2022), ActionFormer (Zhang et al., 2022a), TadML (Deng et al., 2023), TemporalMaxer (Tang et al., 2023), and TriDet (Shi et al., 2023). The comparison results are presented in Table 2, our ASGP-IDet achieves 93.93%, 93.73%, 93.22%, 92.29%, and 87.46% mAP at [0.3:0.7:0.1] tIoU thresholds and achieves an average mAP of 92.13%, which outperforms any other temporal action localisation methods. Specifically, our method outperformed the most competitive ActionFormer approach by 0.49%, 0.95%, 1.46%, 2.59%, and 4.60% in terms of the mAP at [0.3:0.7:0.1] tIoU thresholds.

To gain further insight into the detailed performance, we analysed the detection performance of the proposed ASGP-IDet under different behaviour categories. As shown in Table 3, the proposed ASGP-IDet obtained 96.51%, 91.25% and 88.62% average mAP for mounting, running and fighting, respectively. Our model achieved the best results in terms of the mounting behaviour. Although the performance of running and fighting behaviour was lower than that of mounting behaviour, our method still achieved more than 85% of the average mAP for those two behaviours.

To obtain more insights from the proposed method, the qualitative beef cattle temporal behaviour localisation results for representative videos of different lengths are shown in Fig. 8. Here, we compared the ASGP-IDet method with three top-performing temporal action methods: ActionFormer (Zhang et al., 2022a), TemporalMaxer (Tang et al., 2023), and TriDet (Shi et al., 2023). For example, as shown in Fig. 8(d), our ASGP-IDet located running and fighting behaviour instances accurately in a long nighttime video, whereas the prediction results of ActionFormer, TemporalMaxer and TriDet had significant temporal offsets. This is primarily because ASGP-IDet uses CSEIO Loss to calculate a more detailed instant offset loss. Overall, the results predicted by our ASGP-IDet were the closest to the true temporal label and obtained the minimum temporal instant offsets compared with the other methods.

Table 2

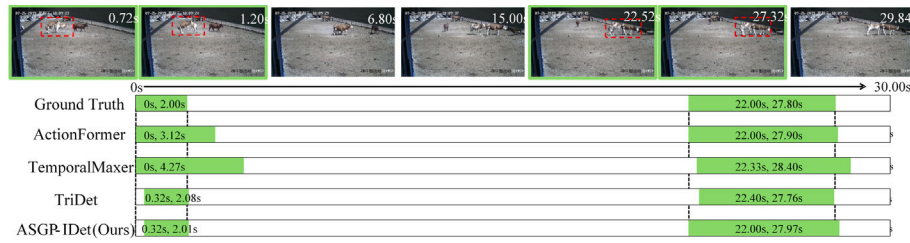
Comparisons with other state-of-the-art temporal action localisation methods on the Cattle Temporal Action dataset.

Methods	mAP@0.3	mAP@0.4	mAP@0.5	mAP@0.6	mAP@0.7	Average mAP
TAGS	83.72%	80.15%	73.80%	63.52%	56.28%	71.49%
ActionFormer	93.44%	92.78%	91.73%	89.70%	82.86%	90.10%
TadML	88.06%	81.10%	76.80%	75.50%	62.80%	76.85%
TemporalMaxer	84.16%	81.87%	80.03%	73.77%	65.28%	77.02%
TriDet	90.78%	90.60%	89.61%	87.11%	83.22%	88.26%
ASGP-IDet(ours)	93.93%	93.73%	93.22%	92.29%	87.46%	92.13%

Table 3

Results for ASGP-IDet with a single behaviour of different categories.

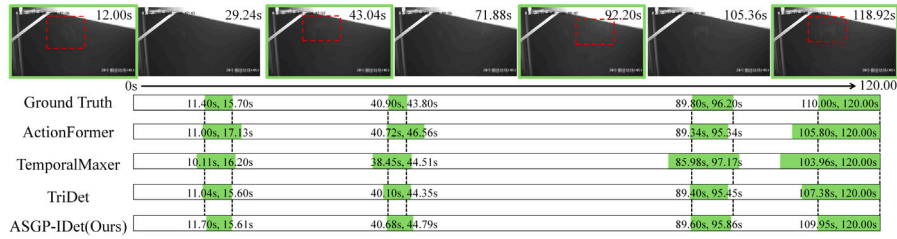
Categories of behaviour	mAP@0.3	mAP@0.4	mAP@0.5	mAP@0.6	mAP@0.7	Average mAP
Mounting	96.83%	96.82%	96.82%	96.82%	95.24%	96.51%
Fighting	91.04%	90.45%	90.40%	90.03%	81.19%	88.62%
Running	93.92%	93.92%	92.43%	90.01%	85.95%	91.25%



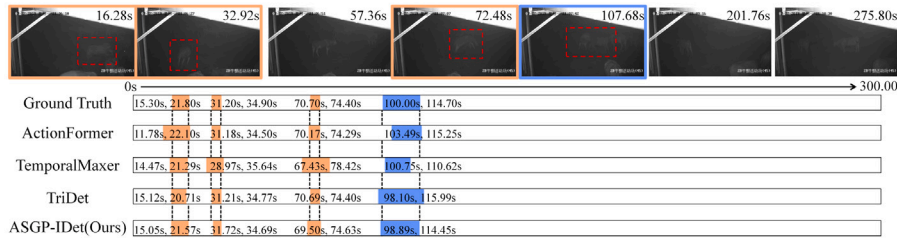
(a) Visualization results of temporal behaviour localisation in a 30 s video



(b) Visualization results of temporal behaviour localisation in a 60 s video



(c) Visualization results of temporal behaviour localisation in a 120 s video



(d) Visualization results of temporal behaviour localisation in a 300 s video

Fig. 8. Visualisation results of different methods for temporal behaviour localisation of beef cattle. A red dashed box marks the area where the behaviour occurs. Green [■] represents mounting behaviour, orange [■] indicates running behaviour, and blue [■] is fighting behaviour. The length of each colour block represents the duration of the behaviour.

Table 4

Timeliness comparisons of different temporal action localisation methods. The best three results are highlighted with red, blue, and green, respectively.

Methods	FLOPs(G)	Params(M)	Inference time (ms)	Average mAP
TAGS	17.84	75.81	135.83	71.49%
ActionFormer	44.91	29.17	226.78	90.10%
TadML	94.71	16.45	201.93	76.85%
TemporalMaxer	23.36	7.09	97.85	77.02%
TriDet	43.33	15.88	166.67	88.26%
ASGP-IDet(ours)	43.34	15.93	92.89	92.13%

Table 5

Ablation study with ASGP module and CSEIO Loss function.

#	Baseline	ASGP module	CSEIO loss	mAP@0.3	mAP@0.4	mAP@0.5	mAP@0.6	mAP@0.7	Average mAP
1	✓			90.78%	90.60%	89.61%	87.11%	83.22%	88.26%
2	✓	✓		94.00%	93.58%	93.19%	90.98%	87.21%	91.79%
3	✓		✓	93.33%	93.12%	92.74%	91.51%	87.17%	91.57%
4	✓	✓	✓	93.93%	93.73%	93.22%	92.29%	87.46%	92.13%

3.2. Analysis of model timeliness

We tested the floating-point operations, parameters, and inference time of the proposed and comparison methods on a test set using an RTX 3090 GPU. From Table 4, it can be seen that the FLOPs and the parameters of our ASGP-IDet are 43.34 G and 15.93 M respectively, both of which achieve the top three performance. The main computational load was that of the ASGP Feature Pyramid. In future studies, we will explore a lighter multiscale feature extraction method to reduce the number of FLOPs and parameters of ASGP-IDet. Specifically, the inference time of ASGP-IDet was 92.89 ms, which was the best performance among all methods. Overall, our method outperformed other state-of-the-art algorithms in the detection performance by a large margin and achieved the fastest detection speed.

3.3. Ablation study

3.3.1. Effectiveness of the ASGP module and CSEIO loss

Our proposed method has two key components: (1) an Aggregate Scalable-Granularity Perception (ASGP) module and (2) a novel centre-start-end instant offset loss (CSEIO Loss). Table 5 lists the effectiveness of these two key components in the test set. Here, TriDet (Shi et al., 2023) was adopted as the baseline algorithm, which did not enable the ASGP module or CSEIO Loss. As shown in Table 5, adding the ASGP module improves the average mAP by approximately 3.53% of the average mAP on baseline #1 (#2 vs. #1). Our CSEIO Loss contributes a total improvement of 3.31% in the average mAP (#3 vs. #1). When further adding CSEIO Loss based on #2, it achieves a 0.34% absolute increase in the average mAP (#4 vs. #2). In general, both the ASGP module and CSEIO Loss are beneficial for beef cattle temporal behaviour localisation tasks. Our full algorithm incorporating all these components (#4) achieves considerable performance gains in terms of the mAP at the [0.3:0.7:0.1] tIoU thresholds when compared to other baseline approaches.

3.3.2. Ablation study on the numbers of ASGP feature pyramid layers

We designed ASGP feature pyramids with two to nine layers and tested their performance on a Cattle Temporal Action dataset. As shown in Table 6, our ASGP feature pyramid with the 6 layers achieved the best result. Considering both the detection accuracy and efficiency, we chose ASGP feature pyramids with six layers in our final model.

3.3.3. Comparisons with different loss function

To verify the effectiveness of the proposed CSEIO Loss, we compared the CSEIO Loss with other loss functions such as Intersection over Union (IoU) Loss (Yu et al., 2016), Generalised Intersection over Union (GIoU) Loss (Rezatofighi et al., 2019), Distance Intersection over Union (DIOU) Loss (Zheng et al., 2019) and Efficient over Union (EIoU) Loss (Zhang

et al., 2022b). As shown in Table 7, our CSEIO Loss outperformed DIOU Loss in most metrics of mAP under different thresholds and obtained the best results with an average mAP of 92.13%, which was 0.34% higher than the most competitive DIOU loss function. The main reason for this is that our CSEIO Loss calculated the temporal offset of the start and end instants for beef cattle behaviour to minimise losses.

3.4. Effect of different interference factors

3.4.1. Effect of different collection times

In real farming scenarios, temporal behaviour labels of surveillance videos must be predicted under different lighting conditions during the day and night. To verify the effectiveness of our method during both daytime and nighttime, we tested our method using daytime and nighttime videos of the test set. As shown in Table 8, our method achieved the best average mAP score of 95.53% during daytime. In addition, our method achieved the best average mAP score of 79.68% in the dark. Although our method may not perform as well in the dark as during the day, it still achieved the best performance in the dark. The main reason for the decrease in performance at night is that weak lighting seriously affects the ability of the camera to capture the movement of beef cattle.

3.4.2. Effect of videos with different lengths

To investigate the effectiveness of our method for videos of different lengths, we tested our method using videos of 30, 60, 120, and 300 s. As listed in Table 9, our method ASGP-IDet achieved the best performance for all video lengths compared to the competitive ActionFormer and TriDet methods. Specifically, for 300 s videos, our method achieved an average mAP score of 78.64%, which was significantly lower than 30, 60, and 120 s videos. The reasons for this phenomenon were that abnormal behaviours in beef cattle were extremely rare, and the proportion of behaviour to background was particularly small in the 300 s videos.

3.4.3. Effect of videos with different numbers of behaviour instances

To further validate the robustness of the proposed ASGP-IDet method, we tested it on videos with different quantities of behaviour instances. As shown in Table 10, ASGP-IDet achieved the best performance in videos containing different quantities of behaviour instances compared with the competitive ActionFormer and TriDet methods. For videos containing four or more instances, our method got 84.92% average mAP, which was lower than that of videos containing one to three behaviour instances. This is because, as the number of behaviour instances increased, the offset of every behaviour instance predicted by ASGP-IDet accumulated, leading to a decrease in the average mAP metric.

Table 6
Analysis of the number of feature pyramid layers.

Number of layers feature pyramid	mAP@0.3	mAP@0.4	mAP@0.5	mAP@0.6	mAP@0.7	Average mAP
2	80.62%	80.02%	76.53%	71.90%	62.92%	74.40%
3	86.65%	85.57%	85.01%	80.41%	75.81%	82.69%
4	91.66%	90.97%	90.91%	89.20%	82.53%	89.05%
5	92.43%	92.23%	92.15%	90.74%	85.43%	90.60%
6	93.93%	93.73%	93.22%	92.29%	87.46%	92.13%
7	93.26%	92.86%	92.32%	90.40%	87.01%	91.17%
8	93.66%	93.52%	92.91%	90.91%	87.36%	91.13%
9	91.69%	91.65%	91.01%	88.75%	82.71%	89.16%

Table 7
Comparisons with different loss functions.

Loss function	mAP@0.3	mAP@0.4	mAP@0.5	mAP@0.6	mAP@0.7	Average mAP
IoU loss	93.65%	93.11%	92.91%	89.52%	85.03%	90.84%
GIoU loss	92.14%	91.83%	91.26%	87.68%	83.05%	89.19%
DIoU loss	94.00%	93.58%	93.19%	90.98%	87.21%	91.79%
EIoU loss	93.57%	93.25%	92.92%	90.66%	86.39%	91.36%
CSEIO loss (ours)	93.93%	93.73%	93.22%	92.29%	87.46%	92.13%

Table 8
Results for ASGP-IDet with different collection times.

Collection times	Methods	Average mAP
Daytime	ActionFormer	95.26%
	TriDet	94.99%
	ASGP-IDet	95.53%
Nighttime	ActionFormer	74.88%
	TriDet	73.38%
	ASGP-IDet	79.68%

Table 9
Results for ASGP-IDet with different lengths of videos.

Lengths of videos	Method	Average mAP
30 s	ActionFormer	85.84%
	TriDet	86.62%
	ASGP-IDet	90.80%
60 s	ActionFormer	91.66%
	TriDet	92.91%
	ASGP-IDet	96.87%
120 s	ActionFormer	90.52%
	TriDet	87.66%
	ASGP-IDet	91.39%
300 s	ActionFormer	75.11%
	TriDet	75.39%
	ASGP-IDet	78.64%

Table 10
Results for ASGP-IDet with different number of behaviour instances in each video.

Number of behaviour instances	Method	Average mAP
1	ActionFormer	90.77%
	TriDet	87.34%
	ASGP-IDet	91.90%
2	ActionFormer	93.77%
	TriDet	91.18%
	ASGP-IDet	94.70%
3	ActionFormer	88.15%
	TriDet	92.43%
	ASGP-IDet	92.08%
4 or more	ActionFormer	82.72%
	TriDet	70.28%
	ASGP-IDet	84.92%

3.4.4. Effect of the area of behaviour

Owing to the different distances between the beef cattle and cameras, the size of the area where behaviour occurs also varied. To investigate the relationship between the size of the behaviour areas and the performance of the temporal behaviour localisation task, we used

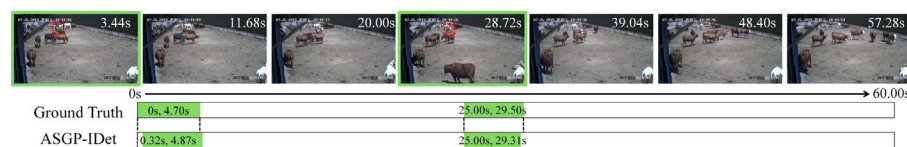
Table 11
Results of ASGP-IDet with different sizes of behaviour area.

Area	Method	Average mAP
Small	ActionFormer	63.68%
	TriDet	65.38%
	ASGP-IDet	76.71%
Medium	ActionFormer	81.97%
	TriDet	78.92%
	ASGP-IDet	83.26%
Large	ActionFormer	90.71%
	TriDet	88.87%
	ASGP-IDet	91.88%

Labelling to annotate the areas in the first frame of each behaviour instance. Subsequently, we categorised these areas as small ($\text{area} \leq 32 \text{ pixels} \times 32 \text{ pixels}$), medium ($32 \text{ pixels} \times 32 \text{ pixels} < \text{area} < 96 \text{ pixels} \times 96 \text{ pixels}$), and large ($\text{area} \geq 96 \text{ pixels} \times 96 \text{ pixels}$) based on the object size criteria in the Microsoft COCO dataset (Lin et al., 2014). According to the statistics, there were 19 behaviour instances with a small area, 317 instances with a medium area, and 75 instances with a large area. The temporal action localisation results of ASGP-IDet for different areas are listed in Table 11. The proposed ASGP-IDet achieved the best average mAP scores of 76.71%, 83.26%, and 91.88% for the small, medium, and large behaviour areas, respectively. The main reason for the decrease in performance in small areas was that the features associated with small areas exhibited limited discriminative power. Furthermore, the visualisation results shown in Fig. 9 have shown that the proposed ASGP-IDet could precisely localise behaviour instances with different areas by benefiting from its ability to extract multi-scale features and aggregate features.

3.5. Limitations and follow-up research directions

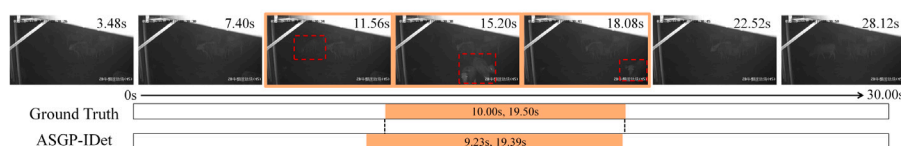
From the above experimental results, it can be seen that our method outperformed other temporal action localisation methods; however, the performance at nighttime needs further improvement. We believe that the reason for this decrease at night was the excessively dark environment. In future work, from the model design perspective, we plan to develop a dark adaptation module to enhance the ability of the model to extract features from nighttime videos. From the data perspective, we will collect thermal infrared videos to improve the accuracy of temporal behaviour localisation under nighttime conditions. In addition, other types of abnormal behaviour, such as lameness, can reflect the health status of beef cattle and are require additional research.



(a) Visualization results of temporal behaviour localisation in the video containing small behaviour area.



(b) Visualization results of temporal behaviour localisation in the video containing medium behaviour area.



(c) Visualization results of temporal behaviour localisation in the video containing large behaviour area.

Fig. 9. Visualisation results of temporal behaviour localisation in video for different areas covering behaviour. A red dashed box marks the area where the behaviour occurs. Green [■] represents mounting behaviour, orange [■] indicates running behaviour, and blue [■] refers fighting behaviour. The length of each colour block represents the duration of the behaviour.

4. Conclusion

Recognising and monitoring beef cattle behaviour are crucial for optimising breeding strategies in precision farming. In this study, we constructed a beef cattle temporal behaviour dataset called Cattle Temporal Action. Using this dataset, we proposed a temporal behaviour localisation method called aggregate scalable-granularity perception instance detection (ASGP-IDet), which integrates a behaviour recognition backbone, a feature pyramid with ASGP blocks, and a multi-branch prediction head to classify behaviour categories and localise behaviour instances. In addition, we introduced a novel temporal distance loss function, the centre-start-end instant offset loss (CSEIO Loss), which works in conjunction with the focal loss to train the ASGP-IDet model. The experimental results on the dataset demonstrated that ASGP-IDet achieved mAP scores of 93.93%, 93.74%, 93.22%, 92.29%, and 87.46% at [0.3:0.7:0.1] tIoU thresholds with an average mAP of 92.13% and an average processing time of 92.9 ms per video. Extensive comparisons with state-of-the-art temporal action localisation methods verified the effectiveness and robustness of the proposed method. Moreover, these experiments confirmed that our method successfully localises the temporal behaviour in beef cattle, offering technical support for monitoring and precision farming.

CRedit authorship contribution statement

Yamin Han: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jie Wu:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Qi Zhang:** Visualization, Software, Investigation, Data curation. **Xilong Feng:** Visualization, Software, Investigation, Data curation. **Yang Xu:** Investigation, Data curation. **Taoping Zhang:** Investigation, Data curation. **Bowen Wang:** Writing – review & editing, Supervision, Resources, Investigation, Conceptualization. **Hongming Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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